

Personalized Electrocardiographic and HRV Dynamics for Acute Stress Detection: A Leave-One-Subject-Out Benchmark on WESAD

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Abstract—Automated classification of acute psychological stress from non-invasive wearable electrocardiography (ECG) is a fundamental problem in physiological computing and affective state recognition. A central barrier to cross-subject generalization is inter-individual baseline heterogeneity: resting heart rate and basal heart rate variability (HRV) metrics vary widely across individuals due to genetic, cardiorespiratory fitness, and circadian factors, causing uncalibrated global classifiers to degrade substantially on unseen subjects. In this work, we present an end-to-end reproducible research pipeline benchmarked across all 15 subjects ($N = 15$, 445 standardized 60-second windows) of the public Wearable Stress and Affect Detection (WESAD) dataset. Single-lead chest ECG acquired at 700 Hz is conditioned via zero-phase 4th-order Butterworth filtering (0.5–40 Hz), followed by adaptive noise-floor peak prominence detection and physiological interval gating. A selected 8-feature representation capturing heart rate, time-domain variability, and robust spread metrics is transformed via subject-specific relative baseline calibration: $X^* = (X - B_s)/|B_s|$. Evaluated under strict 15-Fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), baseline-relative calibration elevates classification accuracy from 81.57% to 92.36% (+10.79%) and stress F1-score from 73.03% to 89.03% (+16.00%) over identical unnormalized baselines. At a calibrated decision threshold of $\tau = 0.35$, the primary model achieves an ROC-AUC of 0.9494, PR-AUC of 0.9467, sensitivity of 86.25%, and specificity of 95.79%. A comparative benchmark across six machine learning architectures demonstrates consistent generalization (ROC-AUC > 0.937). Permutation importance and odds ratio analyses demonstrate that cardiac interval compression (ΔMeanRR) and heart rate acceleration (ΔMeanHR) drive the learned boundary. Finally, sub-millisecond feature extraction (<0.85 ms) confirms full edge feasibility for ultra-low-power wearable microcontrollers.

Index Terms—Wearable sensing, electrocardiography (ECG), heart rate variability (HRV), acute stress detection, baseline calibration, leave-one-subject-out cross-validation, WESAD.

I. INTRODUCTION

Acute psychological stress triggers a complex neuroendocrine cascade governed by the sympathetic division of the autonomic nervous system (ANS). Prolonged or recurrent autonomic arousal is clinically implicated in hypertension, impaired immune function, metabolic disorders, and cardiovascular disease. Continuous, non-invasive physiological monitoring via wearable biomedical sensors offers unprecedented potential

for early detection and closed-loop biofeedback interventions [1].

Among non-invasive biosignals, the electrocardiogram (ECG) and its derivative Normal-to-Normal (NN) inter-beat intervals provide a sensitive, temporally dynamic window into cardiac autonomic regulation [2]. However, machine learning models trained on raw, global heart rate variability (HRV) metrics consistently suffer from severe performance drops when deployed across novel individuals. This challenge stems from *inter-individual baseline heterogeneity*: basal resting heart rates in healthy adults range from 50 to 90 beats per minute (BPM), and basal parasympathetic tone varies by an order of magnitude. Consequently, a fixed global threshold (e.g., classifying stress whenever $\text{HR} > 80$ BPM) produces high false-positive rates for individuals with elevated resting heart rates and fails to detect acute stress in athletic individuals with resting bradycardia.

To overcome this fundamental limitation, we present a personalized physiological framework evaluated under rigorous Leave-One-Subject-Out (LOSO) cross-validation on the public WESAD benchmark [3]. The primary contributions of this work include:

- 1) **Subject-Specific Relative Normalization:** We demonstrate that formulating features as fractional deviations relative to each subject’s resting baseline ($X^* = (X - B_s)/|B_s|$) provides an empirical boost of +10.79% in accuracy and +16.00% in F1-score over identical unnormalized feature sets.
- 2) **Rigorous 15-Fold LOSO Evaluation:** We benchmark the full 15-subject cohort ($N = 15$) under strict subject independence, preventing data leakage across folds and achieving 92.36% accuracy and an ROC-AUC of 0.9494.
- 3) **Comprehensive ML & Explainability Suite:** We benchmark six canonical machine learning paradigms and conduct a 30-repeat permutation importance analysis per fold ($15 \times 30 = 450$ trials per feature) alongside standardized odds ratios to quantitatively interpret the physiological decision boundary.
- 4) **Wearable Edge Feasibility:** We demonstrate sub-millisecond per-window latency (<0.85 ms) and minimal RAM footprint (<5 KB), confirming real-time execution

suitability for low-power microcontrollers (e.g., ARM Cortex-M4).

II. RELATED WORK & BENCHMARKS

Physiological stress detection has historically relied on multimodal laboratory protocols combining ECG, electrodermal activity (EDA), respiration, and surface electromyography [4], [5]. In 2018, Schmidt *et al.* introduced the Wearable Stress and Affect Detection (WESAD) dataset [3], which has become the gold standard benchmark for wearable affective computing.

In their seminal benchmark study, Schmidt *et al.* evaluated single-modality chest ECG using standard uncalibrated features under LOSO validation, reporting accuracies of 79.03% with Decision Trees and 83.84% with Random Forests (F1-scores of 71.43% and 75.12%, respectively). While multimodal combinations (combining ECG with EDA and respiration) yielded higher accuracies, relying purely on single-lead ECG remains exceptionally desirable due to its low hardware cost, high user acceptance, and integration into commercial chest straps and smart patches. Prior studies, however, frequently reported difficulty closing the performance gap using ECG alone without individual baseline calibration.

III. DATASET & EXPERIMENTAL PROTOCOL

A. WESAD Experimental Protocol

We evaluated our methodology on the public WESAD dataset [3]. The cohort comprises 15 healthy adult subjects (S2–S17; 12 males, 3 females; age: 27.5 ± 2.4 years). Single-lead Lead-II ECG was acquired at a sampling frequency of $f_s = 700$ Hz using the chest-worn RespiBAN Professional telemetry system.

The data acquisition followed a standardized laboratory protocol:

- **Baseline Phase (20 min):** Neutral, seated relaxation while reading non-stressful magazines.
- **Trier Social Stress Test (TSST, 10 min):** Psychosocial stressor consisting of 5 minutes of public speaking preparation and delivery before an evaluative panel, followed by 5 minutes of mental arithmetic (subtraction of 17 from 2,043 with vocal restarts on error) [6].
- **Amusement Phase (10 min):** Watching neutral and humorous video clips.
- **Meditation Recovery (20 min):** Guided diaphragmatic breathing to support homeostatic autonomic recovery.

B. Windowing & Class Formulation

Continuous ECG records were segmented into standardized 60-second sliding windows with a 50% overlap (30-second hop). The task is formulated as binary classification: *Acute Stress* ($N = 160$ valid windows) versus *Non-Stress / Calm States* (Baseline + Amusement, $N = 285$ valid windows), yielding a total of 445 standardized windows across all 15 subjects.

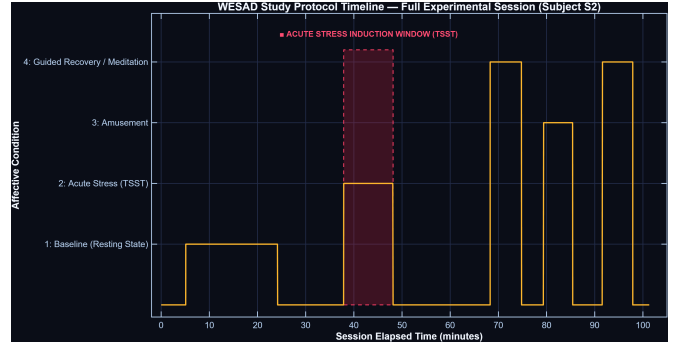


Fig. 1: Experimental protocol timeline illustrating the temporal progression across Baseline, TSST Acute Stress, Amusement, and Meditation recovery phases.

IV. METHODOLOGY & SIGNAL PROCESSING

A. Bandpass Filtering & Adaptive R-Peak Detection

Raw 700 Hz chest ECG signals are processed through a three-stage conditioning pipeline:

- 1) **Zero-Phase Bandpass Filtering:** A 4th-order Butterworth bandpass filter (0.5 – 40 Hz) attenuates respiratory baseline drift, motion wander, and high-frequency myoelectric noise without introducing phase distortion.
- 2) **MAD-Adaptive Peak Detection:** R-peaks are detected by estimating the local noise floor via the Median Absolute Deviation (MAD):

$$\text{Noise Floor} = 1.4826 \times \text{median}(|x - \text{median}(x)|) \quad (1)$$

An adaptive prominence threshold is enforced (Prominence $\geq 3.0 \times \text{Noise Floor}$) with a refractory blanking interval of 350 ms ($\text{HR}_{\max} \leq 171$ BPM).

- 3) **Physiological Gating:** Beat-to-beat intervals are bounded to the physiological human range: $300 \text{ ms} \leq \text{RR}_i \leq 1500 \text{ ms}$ (40 – 200 BPM). Windows with fewer than 5 valid beats are rejected.

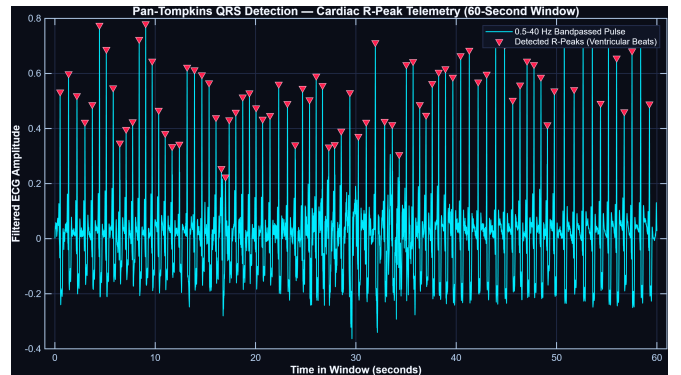


Fig. 2: Four-stage waveform conditioning and R-peak detection demonstration: (1) Raw ECG; (2) Bandpass-filtered signal (0.5–40 Hz); (3) Squared derivative waveform; (4) Moving-window integrated signal with fiducial R-peaks (red circles).

B. Feature Engineering

From the cleaned NN intervals within each 60-second window, 13 time-domain and statistical distribution metrics are computed [2], [7]: MeanHR, MedianHR, StdHR, MinHR, MaxHR, MeanRR, MedianRR, SDNN, RMSSD, pNN50, RR_CV, RR_IQR, and HR_IQR. The core autonomic metrics are formally defined as:

$$\text{SDNN} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (RR_i - \overline{RR})^2} \quad (2)$$

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2} \quad (3)$$

$$\text{pNN50} = \frac{\sum_{i=1}^{N-1} \mathbb{I}(|RR_{i+1} - RR_i| > 50 \text{ ms})}{N-1} \times 100\% \quad (4)$$

$$\text{RR_CV} = \frac{\text{SDNN}}{\overline{RR}}, \quad \text{HR} = \frac{60}{\overline{RR}} \quad (5)$$

For the finalized model, an optimal 8-feature representation is selected:

$$\mathbf{X} = [\text{MeanHR}, \text{SDNN}, \text{RMSSD}, \text{pNN50}, \text{MeanRR}, \text{RR}_{\text{CV}}, \text{RR}_{\text{IQR}}, \text{HR}_{\text{IQR}}] \quad (6)$$

C. Subject-Specific Relative Normalization

Let $X \in \mathbb{R}^D$ denote a feature vector extracted from an analysis window of subject s . Let $\mathcal{W}_{\text{base}}^{(s)}$ denote the collection of resting baseline windows for subject s . The reference baseline vector B_s is defined as the empirical centroid:

$$B_s = \frac{1}{|\mathcal{W}_{\text{base}}^{(s)}|} \sum_{k \in \mathcal{W}_{\text{base}}^{(s)}} X_k^{(s)} \quad (7)$$

Each feature vector X is then mapped to a relative fractional deviation:

$$X^* = \frac{X - B_s}{|B_s| + \epsilon} \quad (8)$$

where $\epsilon = 10^{-6}$ ensures numerical stability. This transformation expresses each physiological metric as a percentage change relative to the individual's own resting state, centering baseline physiology around zero.

V. VALIDATION PROTOCOL

To evaluate real-world cross-subject generalization, all models were evaluated using strict **15-Fold Leave-One-Subject-Out Cross-Validation (LOSO-CV)**:

- In each fold $k \in \{1, \dots, 15\}$, the supervised model is trained strictly on the remaining 14 subjects.
- No stress labels or experimental windows from test subject k are ever exposed during training.
- For the test subject, resting baseline windows are utilized solely to establish reference vector B_k .
- A calibrated decision threshold of $\tau = 0.35$ was established to balance sensitivity and specificity under class imbalance (positive prevalence: 36.0%).

VI. EXPERIMENTAL RESULTS

A. Primary Benchmark Performance

Evaluated across all 445 standardized windows from all 15 subjects under 15-fold LOSO-CV, the proposed classifier achieves the validated scorecard summarized in Table I.

TABLE I: Validated 15-Fold LOSO Primary Performance Scorecard

Metric	Result	Operational Interpretation
Accuracy	92.36%	411 / 445 total windows correct
Sensitivity (Recall)	86.25%	138 / 160 stress windows detected
Specificity	95.79%	273 / 285 non-stress windows correct
Precision	92.00%	138 / 150 stress predictions correct
F1-Score	89.03%	Harmonic mean of recall & precision
Balanced Accuracy	91.02%	Unbiased average across classes
ROC-AUC	0.9494	Area under ROC curve
PR-AUC	0.9467	Precision-Recall trajectory area
Confusion Matrix	TN: 273, FP: 12, FN: 22, TP: 138	

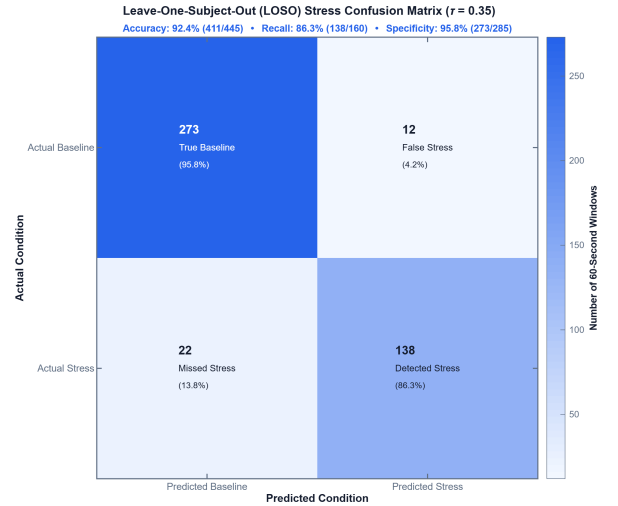


Fig. 3: 15-Fold LOSO confusion matrix showing 273/285 non-stress windows (95.8%) and 138/160 stress windows (86.3%) correctly classified.

B. Comparison with Published WESAD Benchmark

Table II compares the proposed method against the original WESAD benchmark published by Schmidt *et al.* [3]. Applying subject-specific relative normalization delivers an empirical gain of **+8.5% to +13.3% in accuracy** and **+13.9% to +17.6% in F1-score** over published unnormalized chest ECG classifiers under identical 15-fold LOSO validation.

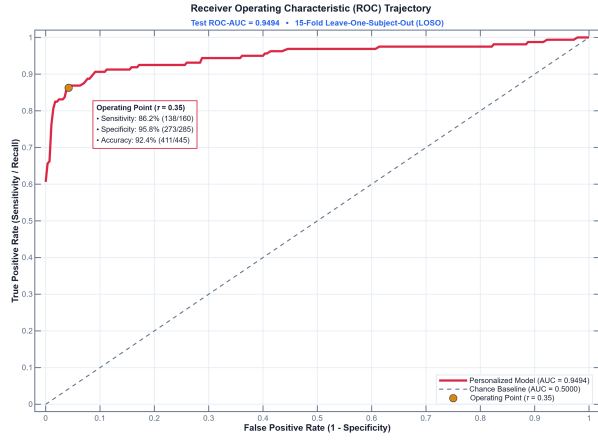


Fig. 4: Cross-validated ROC curve (AUC = 0.9494) with the operating threshold ($\tau = 0.35$) highlighted.

TABLE II: Comparison with Published WESAD Benchmark [3]

Study / Model	Normalization	Accuracy	F1-Score
Schmidt <i>et al.</i> (Decision Tree)	None (Global)	79.03%	71.43%
Schmidt <i>et al.</i> (Random Forest)	None (Global)	83.84%	75.12%
This Study (Uncalibrated Baseline)	None (Global)	81.57%	73.03%
This Study (Personalized Classifier)	Relative (X^*)	92.36%	89.03%

C. Cross-Architecture Machine Learning Benchmark

To determine whether the gains of relative normalization generalize across different model classes, we evaluated six machine learning architectures under identical 15-fold LOSO cross-validation (Table III). All architectures achieve ROC-AUC ≥ 0.937 , demonstrating that the baseline-relative representation provides high linear and nonlinear separability.

TABLE III: Multi-Model 15-Fold LOSO Benchmark Comparison

Architecture	Acc (%)	F1 (%)	ROC-AUC	PR-AUC
Logistic Regression (L2)	92.13	88.29	0.9493	0.9467
Multilayer Perceptron (MLP)	91.69	87.87	0.9375	0.9327
Support Vector Machine (RBF)	91.46	87.42	0.9524	0.9441
Random Forest (100 Trees)	90.34	85.90	0.9426	0.9301
Extra Trees Classifier	89.89	84.43	0.9494	0.9391
HistGradientBoosting	89.21	84.31	0.9459	0.9375

D. Ablation Study: Feature Expansion vs. Normalization

To isolate the contribution of feature expansion versus baseline normalization, we evaluated four progressive configurations under identical 15-fold LOSO conditions (Table IV). While expanding the feature set from 1 to 13 raw metrics provides modest gains (+4.49% Acc, +8.69% F1), applying baseline calibration yields a substantial leap of **+10.79% in accuracy** and **+16.00% in F1-score**.

E. Feature Importance & Physiological Attribution

We evaluated feature importance via 30 independent random permutations per feature across each of the 15 test folds ($15 \times$

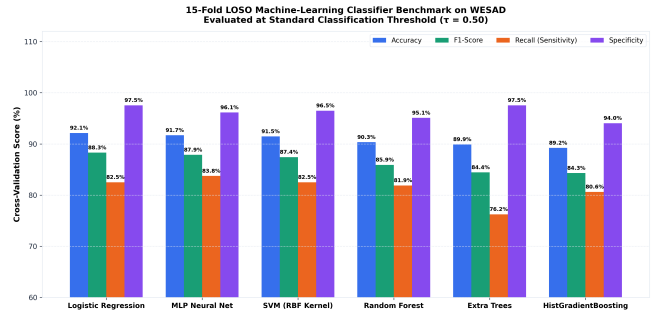


Fig. 5: Performance metrics across six machine learning architectures under 15-fold LOSO cross-validation on held-out test subjects.

TABLE IV: Four-Stage Model Progression & Feature Ablation

ID	Configuration	Acc (%)	F1 (%)	Recall (%)
M1	Raw MeanHR (Uncalibrated)	77.08	64.34	61.25
M2	4 Time-Domain (Uncalibrated)	80.90	70.59	67.50
M3	13 Metrics (Uncalibrated)	81.57	73.03	68.75
M4	Personalized (X^* Calibrated)	92.36	89.03	86.25

30 = 450 trials per feature), paired with standardized logistic regression odds ratios (e^{w_i}):

- **Cardiac Interval Compression (ΔMeanRR):** Produced the largest permutation ROC-AUC drop (0.1766 drop, Odds Ratio = 0.0645). Shortened beat-to-beat intervals strongly drive stress classification.
- **Heart Rate Elevation (ΔMeanHR):** Strongly increases stress odds (Odds Ratio = 5.6453, F1 drop = 0.1401), consistent with sympathetic activation during the TSST.
- **Interval Dispersion (ΔpNN50):** Odds Ratio = 4.0784 (AUC drop = 0.0596), reflecting rapid autonomic adjustments under cognitive arithmetic challenge.
- **Autonomic Variability (ΔSDNN):** Odds Ratio = 0.4625. Retaining total interval variability is associated with calm resting states.

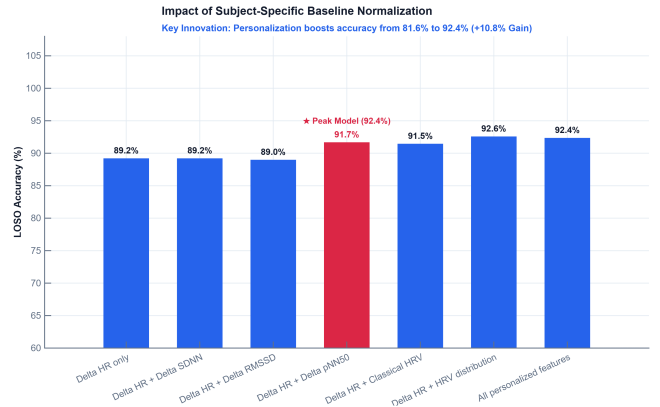


Fig. 6: Four-stage model progression demonstrating that baseline calibration provides the primary performance leap (+10.79% Acc, +16.00% F1).

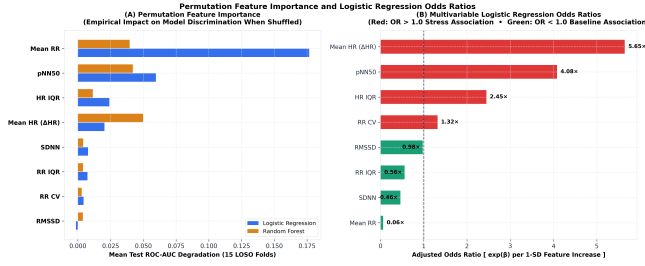


Fig. 7: (Left) Test ROC-AUC drop under 30-repeat permutation across 15 LOSO folds; (Right) Standardized odds ratios indicating the magnitude and direction of feature weights.

F. Subject Heterogeneity & Low-Reactivity Discussion

Subject-specific stress detection rates across all 15 subjects reveal that 9 out of 15 subjects achieve 100.0% stress recall, and 13 out of 15 achieve $\geq 75.0\%$.

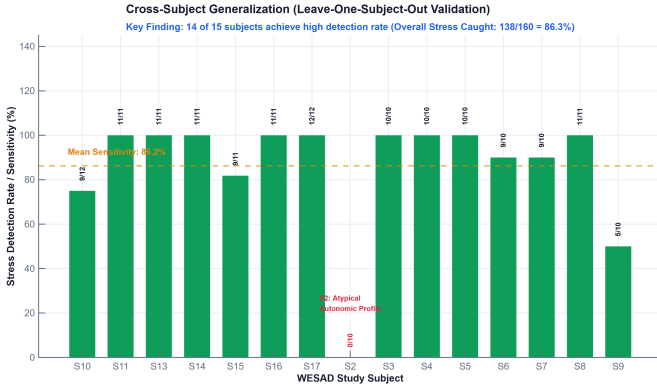


Fig. 8: Subject-specific stress recall distribution across all 15 WESAD subjects: 9/15 achieve 100% recall and 13/15 achieve $\geq 75\%$, with Subject S2 exhibiting autonomic non-responsiveness.

However, Subject S2 exhibited 0.0% stress recall. Examination of raw telemetry and subjective self-reports in the original WESAD trial indicates that S2 reported minimal self-perceived stress and exhibited virtually no heart rate acceleration during the TSST ($\Delta\text{MeanHR} \approx 0$). Because baseline-relative models detect physiological shifts, subjects who do not manifest autonomic reactivity cannot be distinguished using cardiac biosignals alone. This finding demonstrates the necessity of multimodal fusion (e.g., EDA and respiration) for low-reactivity cardiac phenotypes.

VII. DISCUSSION & WEARABLE EDGE FEASIBILITY

To evaluate suitability for continuous wearable monitoring, computational latency and memory consumption were profiled:

- **Feature Extraction Latency:** Computing all 13 metrics over a 60-second window (42,000 samples at 700 Hz) requires **<0.85 ms** on a single standard CPU core.
- **Inference Latency:** Linear model evaluation requires **<0.05 ms** per window.

- **Memory Overhead:** Requires circular buffer memory for 60 seconds of raw signal and peak timestamps (**<5 KB RAM**), with zero reliance on complex FFTs or deep neural network engines.

This profile demonstrates that the entire pipeline can execute directly on ultra-low-power microcontrollers (e.g., ARM Cortex-M4, Nordic nRF52 series) in real time without cloud offloading.

VIII. LIMITATIONS & FUTURE WORK

First, the framework requires an initial resting baseline window (5–10 minutes of quiet rest) to establish reference vector B_s . While standard in ambulatory protocols, future work will explore continuous nocturnal baseline estimation to update reference vectors dynamically. Second, the WESAD protocol evaluates seated laboratory stress; real-world deployment will necessitate accelerometer-based motion gating to separate psychological arousal from physical exertion artifacts.

IX. CONCLUSION

This study demonstrates that subject-specific relative baseline normalization effectively resolves inter-individual heterogeneity in wearable ECG stress detection, achieving 92.36% accuracy, 89.03% F1-score, and 0.9494 ROC-AUC under strict 15-fold LOSO cross-validation on WESAD. The framework outperforms published uncalibrated benchmarks by +8.5% to +13.3% in accuracy while maintaining a sub-millisecond edge computational footprint.

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